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STM32U5 Keyed RDP

STM32U5 Keyed RDP

Introduction

- All the STM32 integrate security mechanism to protect the embedded flash content : the readout protection aka RDP.
- After a reminder about RDP protection and its associated life cycle, we will present the new keyed RDP mode introduced in the STM32U5
- This new mode allows to control all the possible RDP level regression in a secure way.

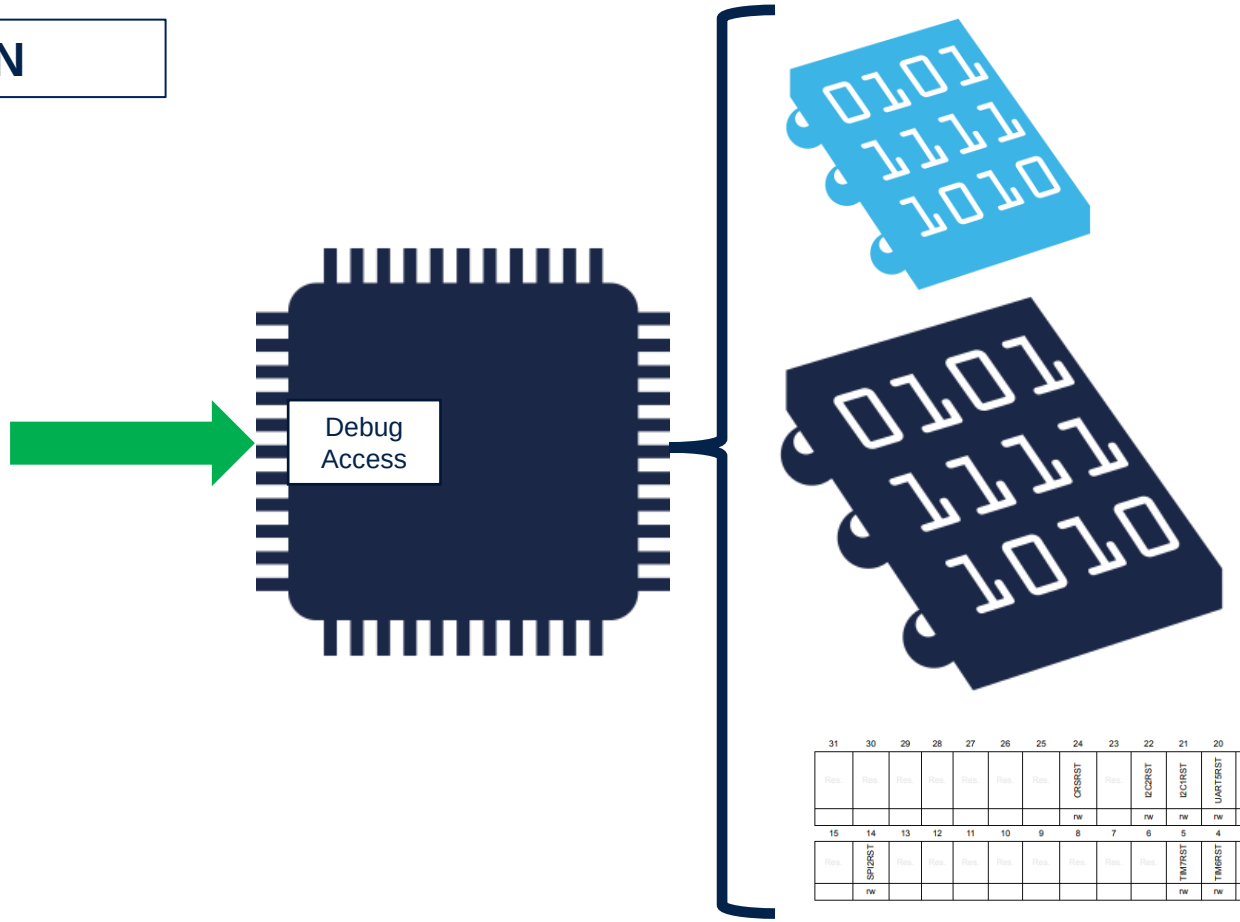
Reminder about RDP protection

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Reminder about RDP protection

RDP level 0 without TrustZone

Device OPEN



SRAM
SRAM1
SRAM2
SRAM3

FLASH
System Flash
- Option byte modification
- Embedded Bootloader

User Flash

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	CRSRST	Res.	I2C2RST	I2C1RST	UART5RST	UART4RST	USART3RST	USART2RST	Res.
							rw		rw	rw	rw	rw	rw	rw	

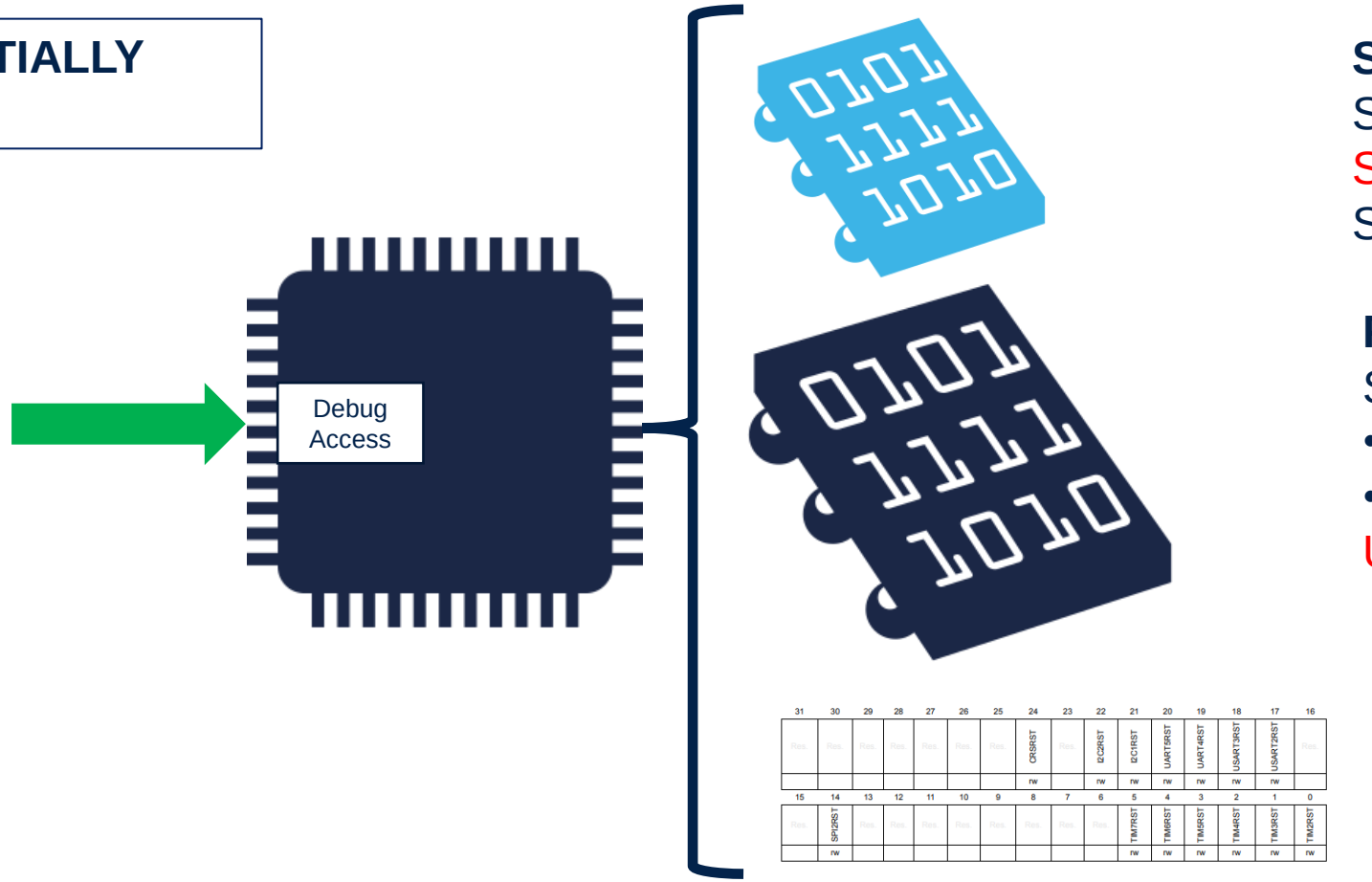
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SPI2RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST
	rw									rw	rw	rw	rw	rw	rw

IPs register
- I2C/USART/DMA….

Reminder about RDP protection

RDP level 1 without TrustZone

Device PARTIALLY CLOSED



SRAM
SRAM1
SRAM2
SRAM3

FLASH
System Flash
- Option byte modification
- Embedded Bootloader

User Flash

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	CRSRST	Res.	I2C2RST	I2C1RST	UART5RST	UART4RST	USART3RST	USART2RST	Res.
							rw		rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SPI2RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST
	rw									rw	rw	rw	rw	rw	rw

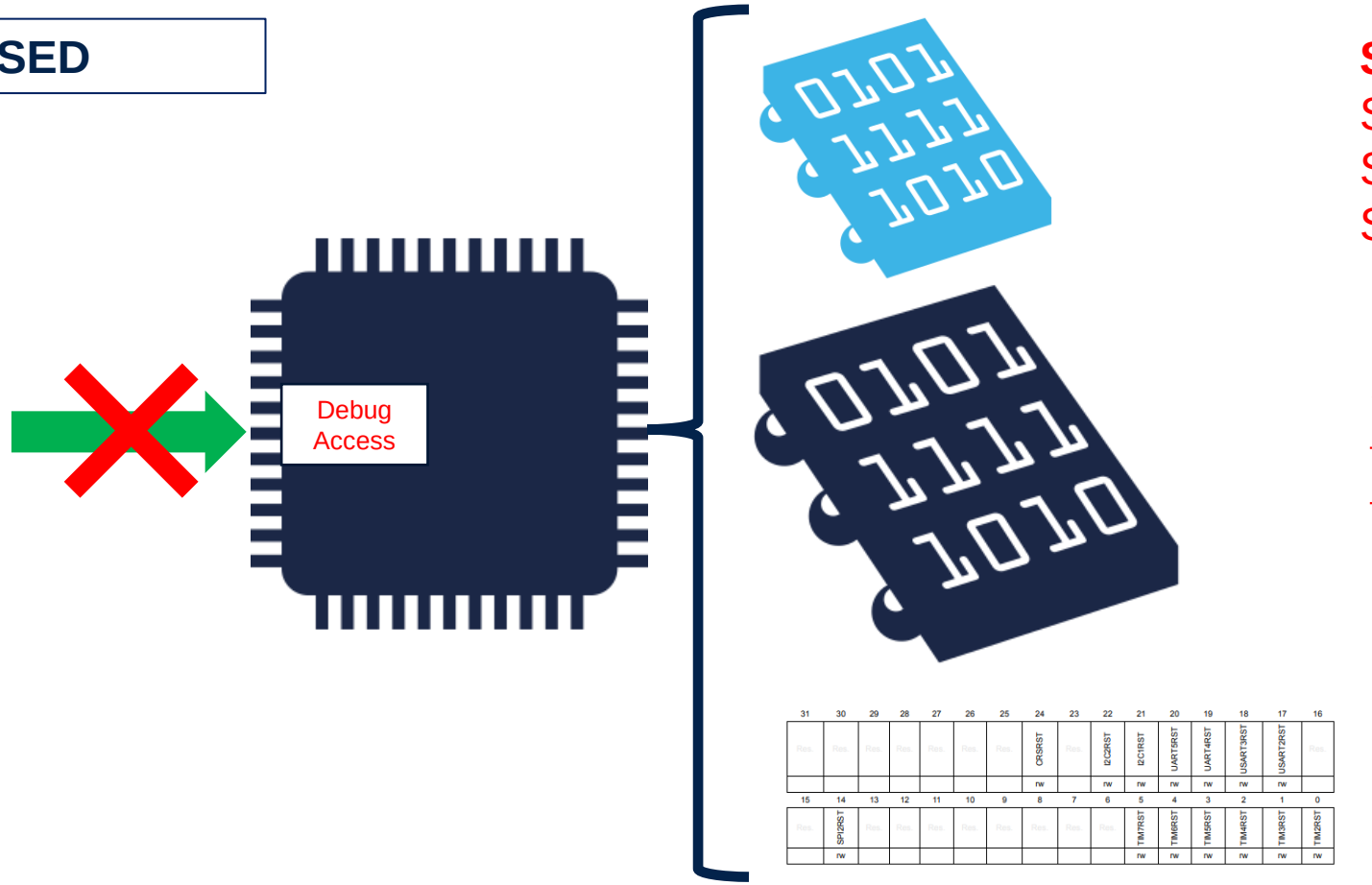
IPs register
- I2C/USART/DMA….

feature : not accessible through debugging link

Reminder about RDP protection

RDP level 2 without TrustZone

Device CLOSED



SRAM
SRAM1
SRAM2
SRAM3

FLASH
System Flash
- ~~Option byte modification*~~
- ~~Embedded Bootloader*~~

User Flash

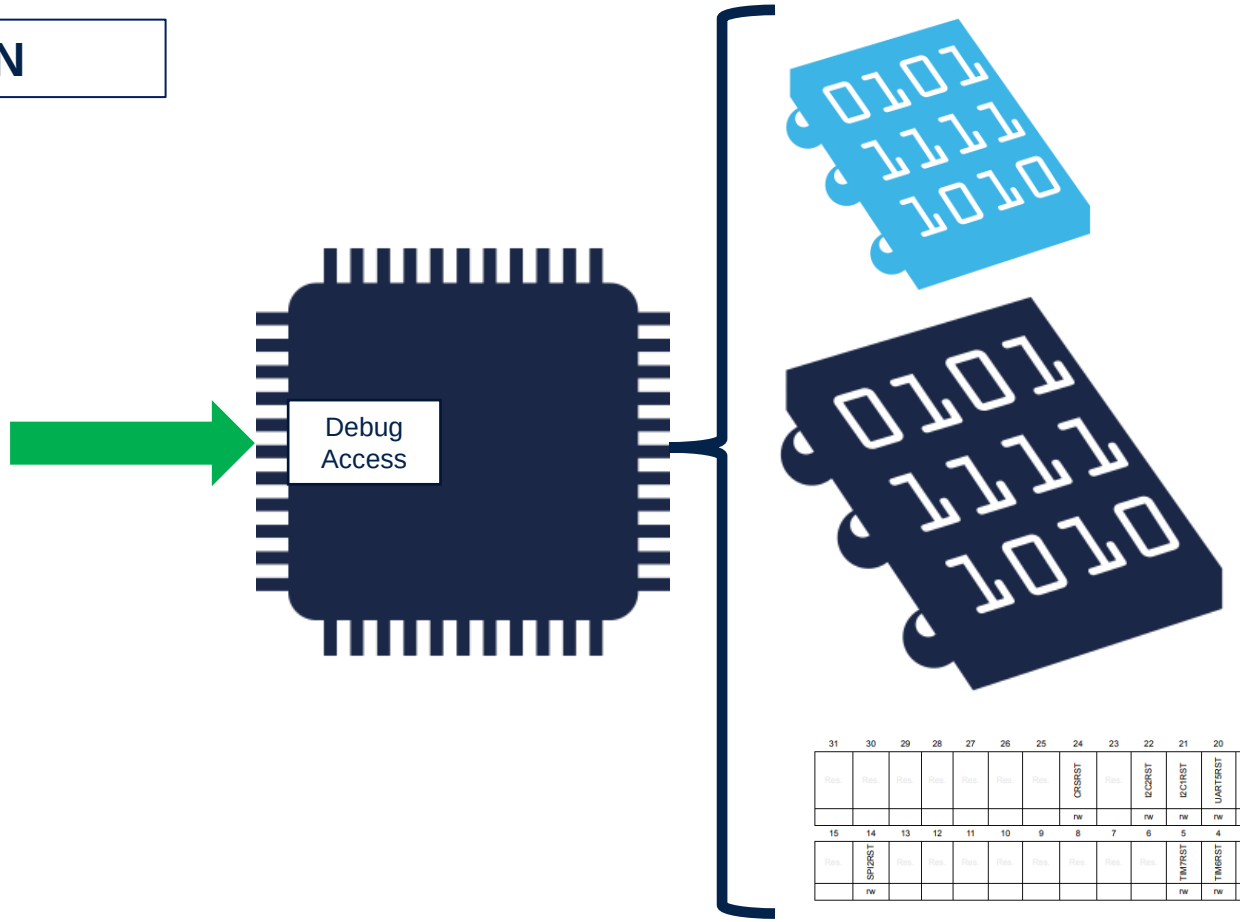
IPs register
- I2C/USART/DMA….

feature : not accessible through debugging link
~~feature~~* : deactivated

Reminder about RDP protection

RDP level 0 with TrustZone

Device OPEN



SRAM
Secure SRAM
Unsecure SRAM

FLASH
System Flash
- Option byte modification
- Embedded Bootloader

Secure User Flash
Unsecure User Flash

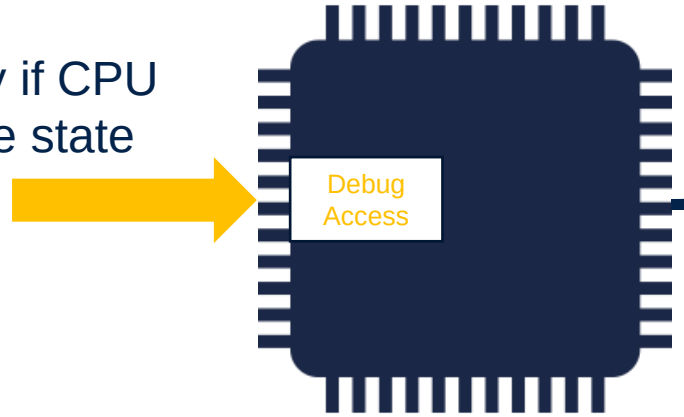
IPs register
- Secure IPs register
- Unsecure IPs register

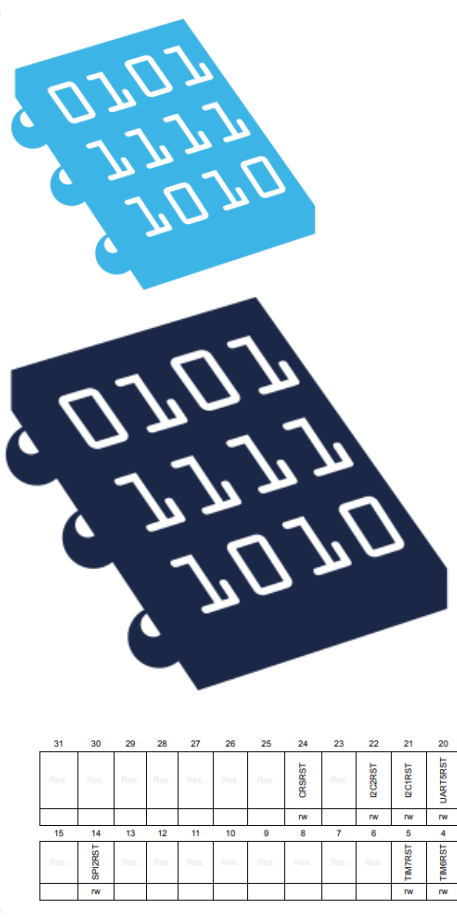
Reminder about RDP protection

RDP level 0.5 with TrustZone

Device PARTIALLY CLOSED

Connect only if CPU in non-secure state





SRAM

Secure SRAM

Unsecure SRAM

FLASH

System Flash

- Option byte modification
- Embedded Bootloader

Secure User Flash

Unsecure User Flash

IPs register

- Secure IPs register
- Unsecure IPs register

feature : not accessible through debugging link

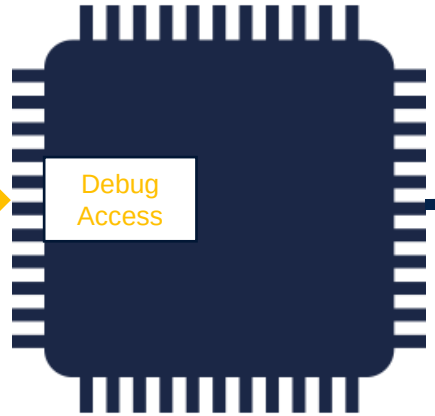
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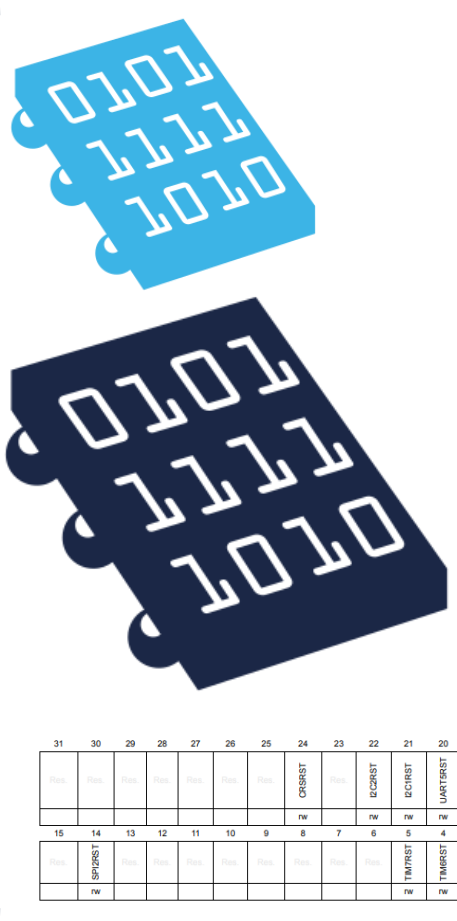
Reminder about RDP protection

RDP level 1 with TrustZone

Device PARTIALLY CLOSED

Connect only if CPU in non-secure state





SRAM

Secure SRAM

Unsecure SRAM

FLASH

System Flash

- Option byte modification
- Embedded Bootloader

Secure User Flash

Unsecure User Flash

IPs register

- Secure IPs register
- Unsecure IPs register

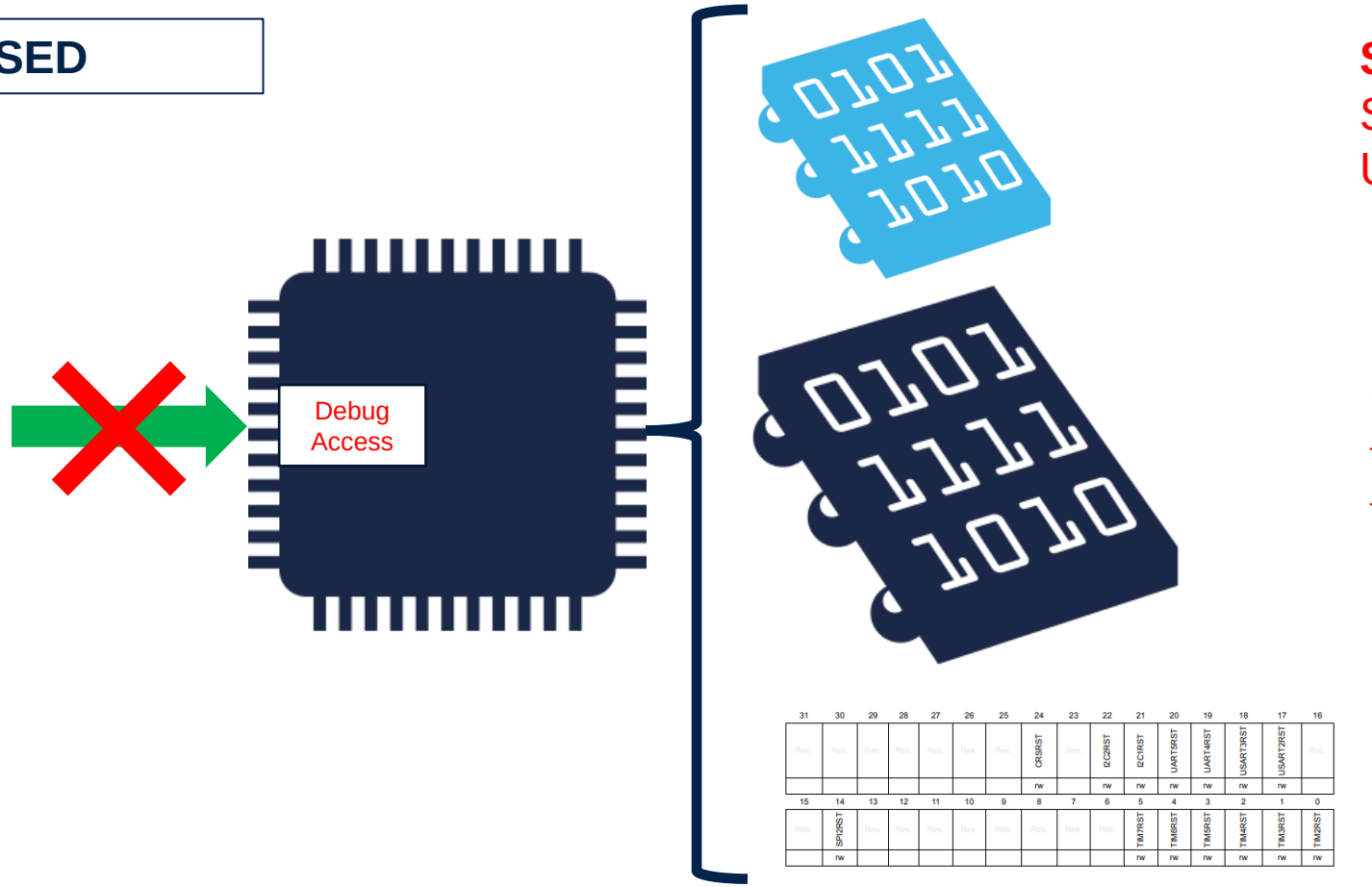
feature : not accessible through debugging link

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Reminder about RDP protection

RDP level 2 with TrustZone

Device CLOSED



SRAM

Secure SRAM

Unsecure SRAM

FLASH

System Flash

- ~~Option byte modification~~
- ~~Embedded Bootloader~~

Secure User Flash

Unsecure User Flash

IPs register

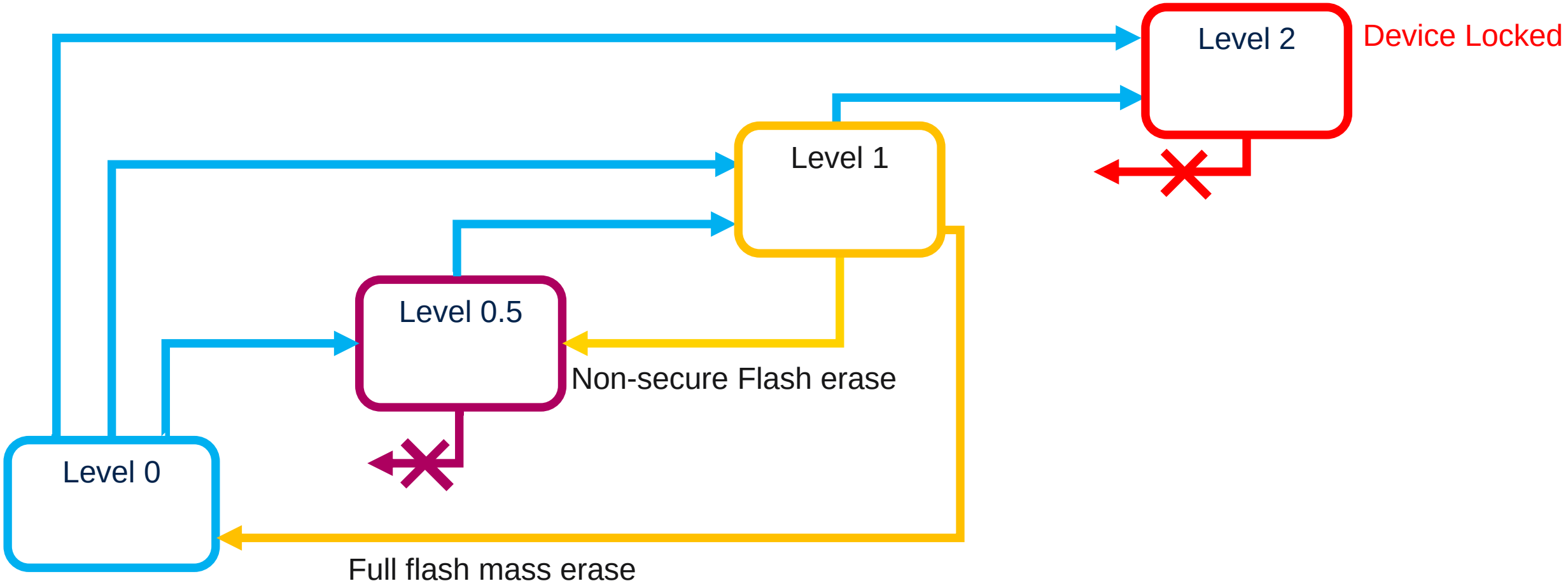
- Secure IPs register
- Unsecure IPs register

feature : not accessible through debugging link

~~feature~~* : deactivated

Reminder about RDP protection

RDP transitions



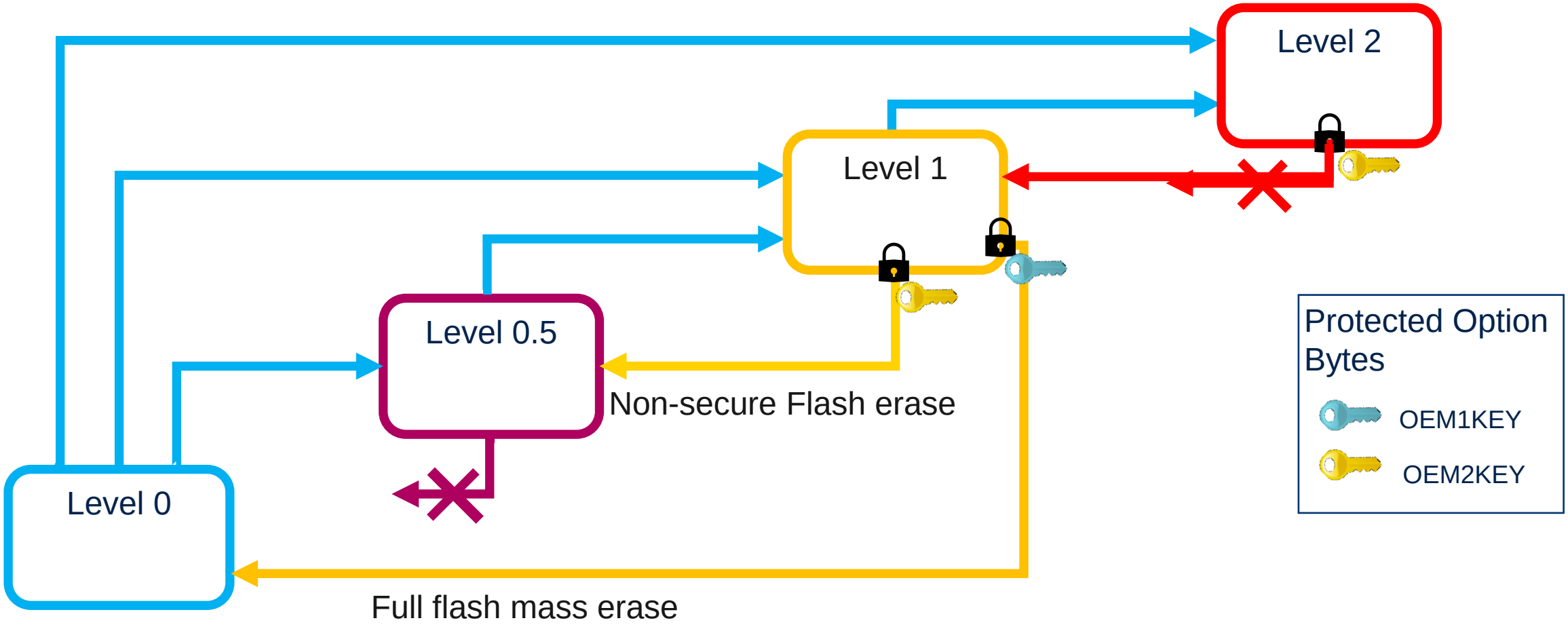
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U5 Keyed RDP

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RDP transitions



Keyed RDP regression

- RDP Password mode
  - Password-based RDP level **regression**
  - Support two different OEMKEY password keys:

OEM1KEY : RDP level regression from **Level 1** to **Level 0**

OEM2KEY : RDP level regression from **Level 2** to **Level 1** and **Level 1** to **Level 0.5**

Note: **Getting the password will never compromise data confidentiality, it allows only to regress the RDP protection.**

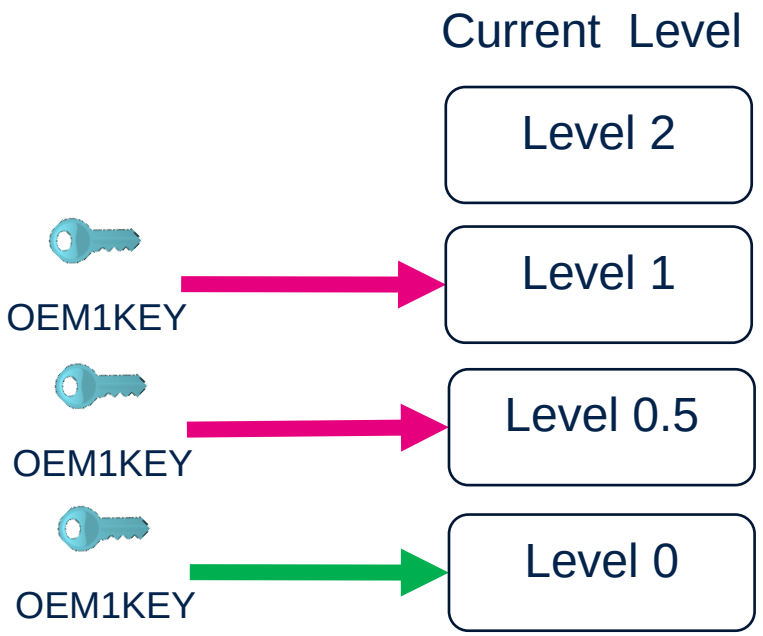
Key handling

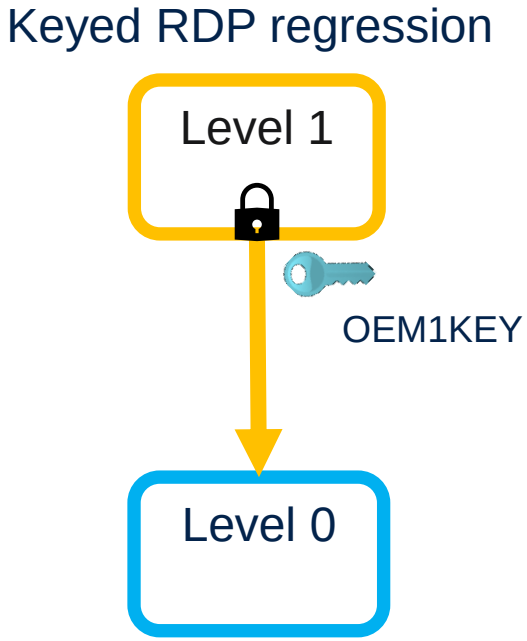
OEM1KEY/OEM2KEY

- OEMxKEY are stored in special protected options bytes in flash system memory.
- When the OEMxKEY are provisioned
  OEMxLOCK bits are set in FLASH NSSR register.
- OEMxKEY option bytes are write only. There is no way to read them…

Key handling

OEM1KEY provisioning regarding RDP level





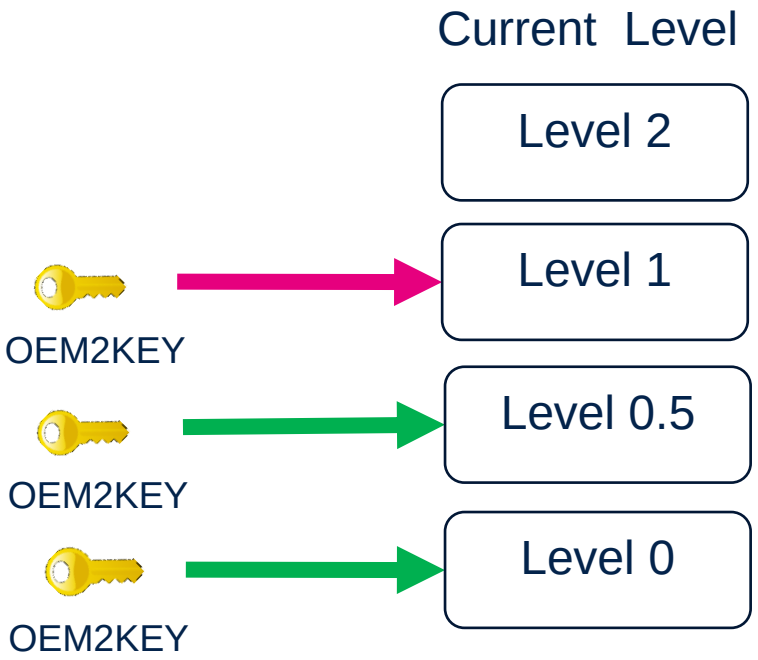
Key can be provisioned, erased and modified

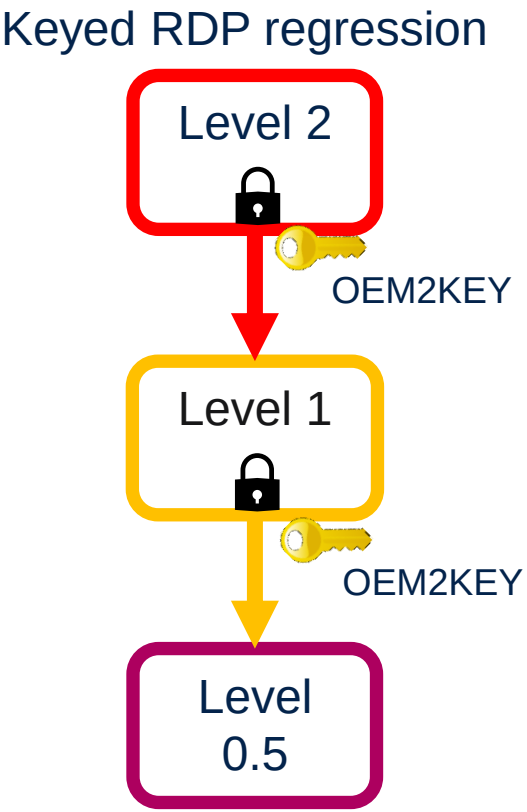
Key can be provisioned

Note: Key is erased if its value is set to 0xffffffff-0xffffffff
if key is erased or not initialized, RDP legacy mode apply

Key handling

OEM2KEY provisioning regarding RDP level





Key can be provisioned, erased and modified

Key can be provisioned

Note: Key is erased if its value is set to 0xffffffff-0xffffffff
if key is erased or not initialized, RDP legacy mode apply

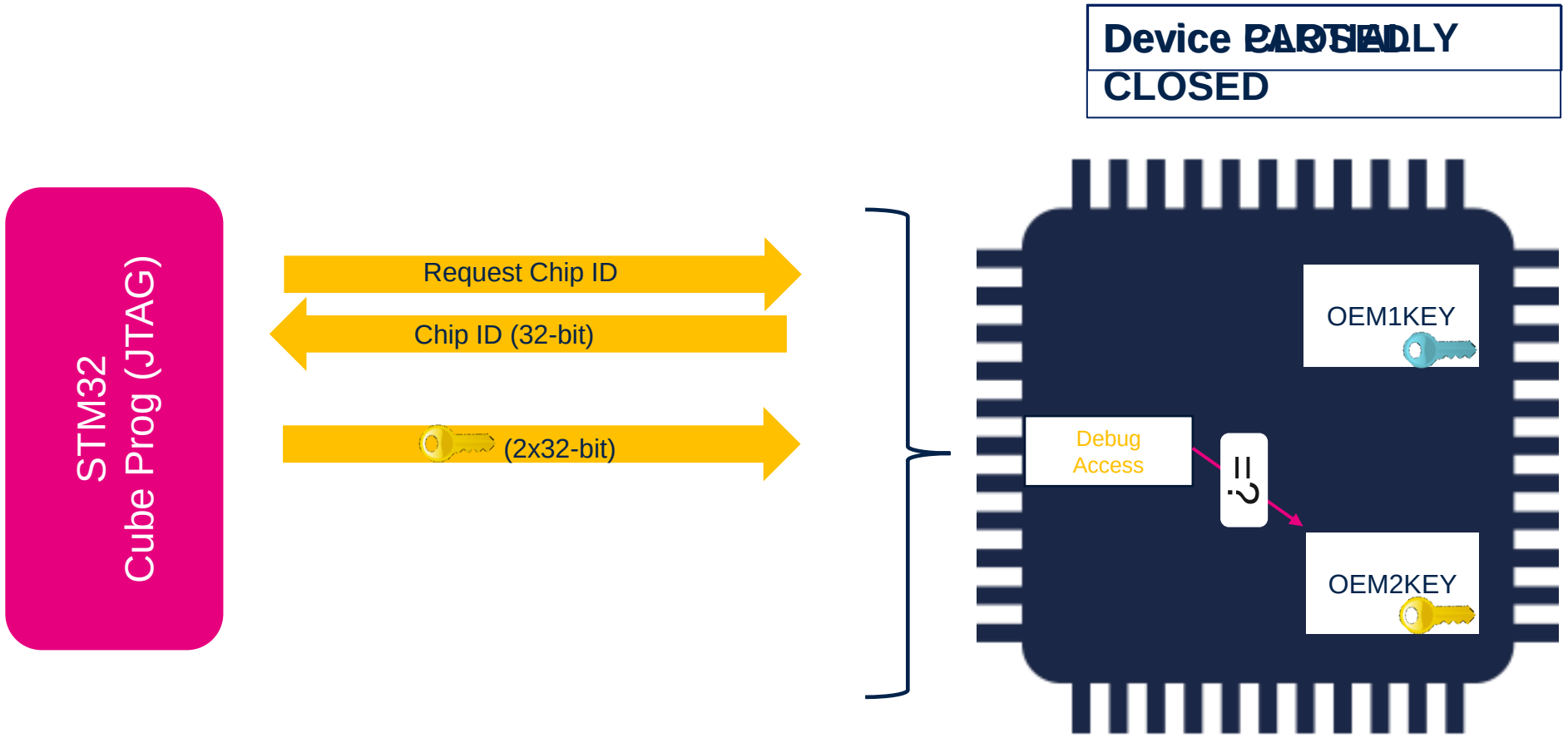
Key management

The 32-bit device specific ID

- Device authentication ID : 32 bits unique device ID provisioned by ST
- Reading through JTAG (except in RDP level2 Legacy mode) from any RDP state
- The Device ID can be used
  - to parse a database to find the corresponding key
  - as part of a specific cryptographic scheme to find OEM1KEY and OEM2KEY.

Keyed regression

RDP level 2 to RDP 1



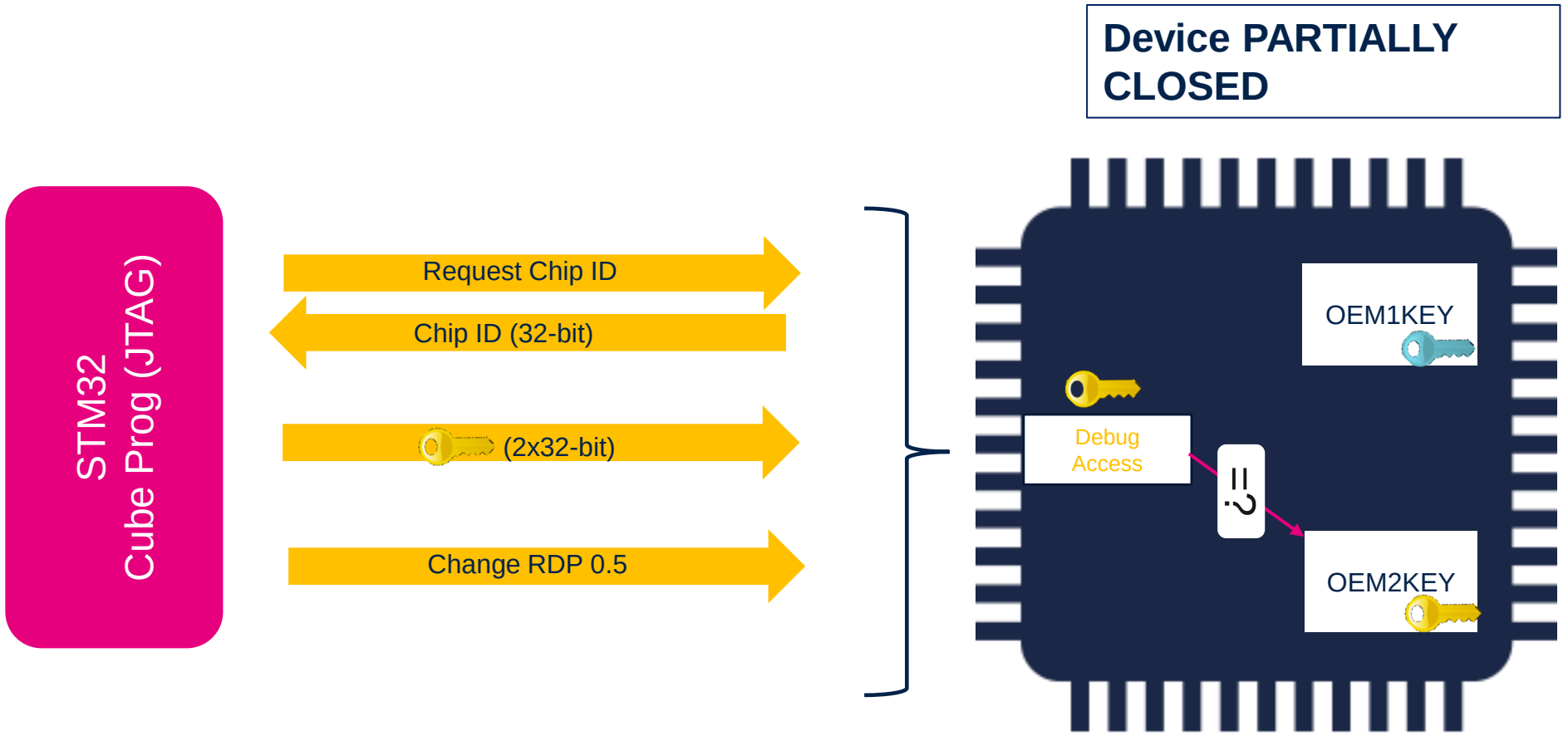
RDP 1 + POR

Note: in case of mismatch key value, POR is needed for a new trial

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Keyed regression

RDP level 1 to RDP 0.5



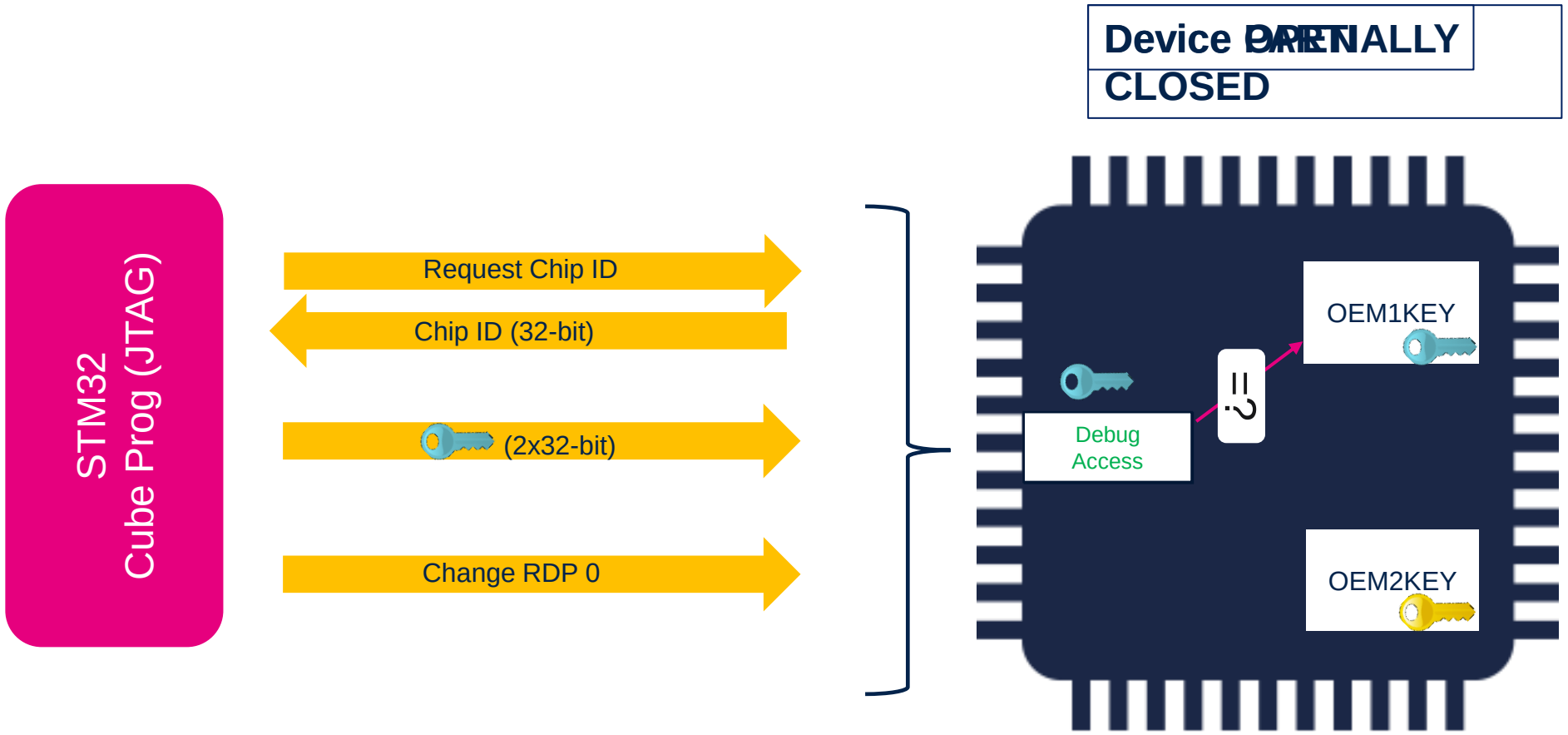
Note: in case of mismatch key value, POR is needed for a new trial

RDP 0.5 - Non secure Flash erased

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Keyed regression

RDP level 1 to RDP 0



Note: in case of mismatch key value, POR is needed for a new trial

RDP 0 - Flash Mass erased

U5 RDP legacy mode

Good practice

- If you plan to rely on RDP legacy mode and the chip has been provisioned before you get it, you may lock the device….
- Good practice before any RDP modification :
  - check if some OEMxKEYS has been provisioned thanks OEMxLOCK bits in the FLASH NSSR register
  - Erase them by programming keys with 0xFFFFFFFF 0xFFFFFFFF

Useful documentation

U5 keyed RDP

- RM0456 : Reference manual STM32U575/585 Arm®-based 32-bit MCUs
- AN5347 : Arm® TrustZone® features for STM32L5 and STM32U5 Series

U5 keyed RDP

Conclusion

- STM32U5 still have the RDP protection mechanism with legacy lifecycle.
- It brings an improvement of this mechanism with the keyed feature.
- This mechanism allows you to control, in a secure way, RDP regression from any RDP state (include RDP level 2).

Now it’s time to experiment this !

Our technology starts with You



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